

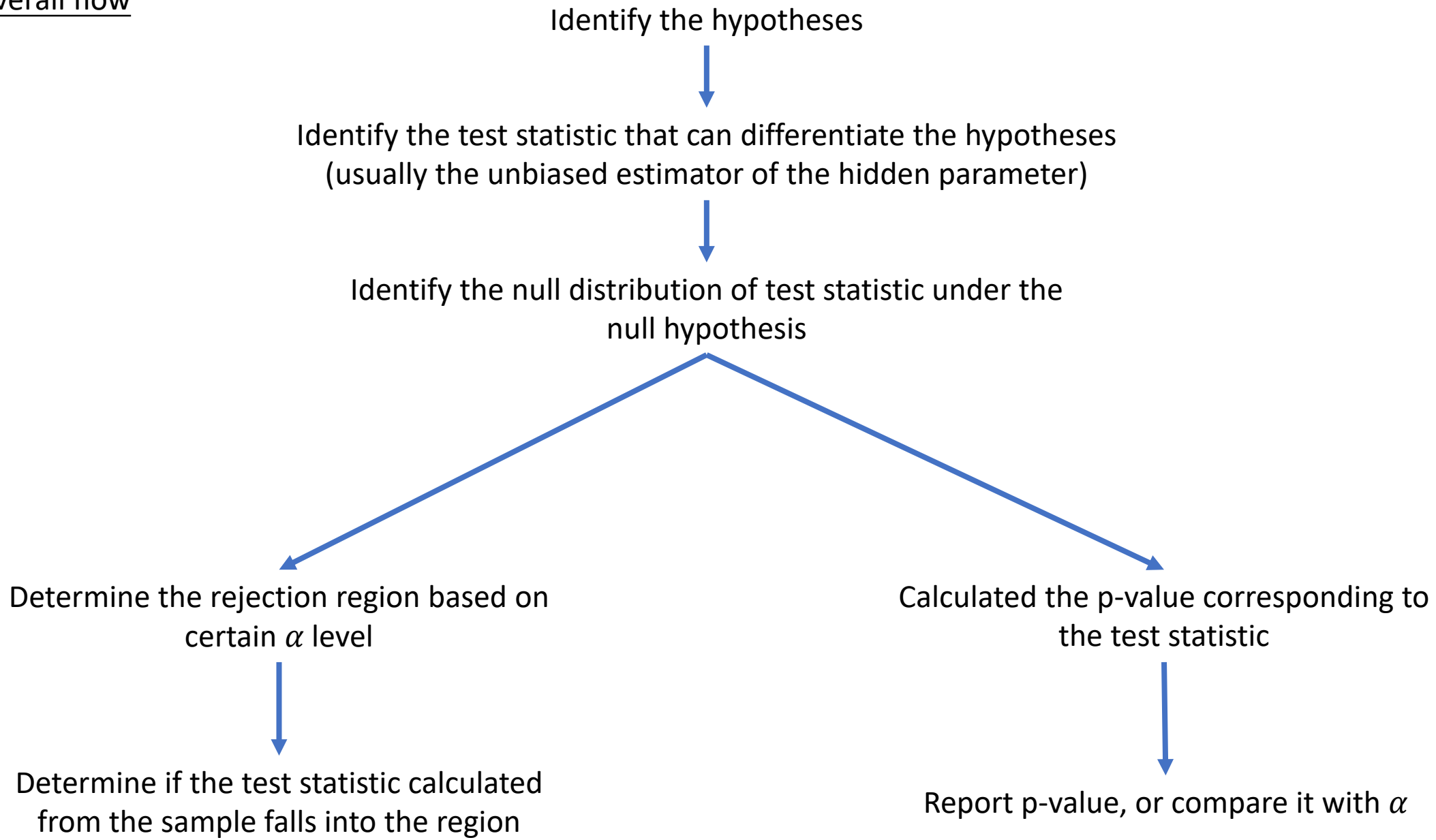
STA 2002

Lecture 8

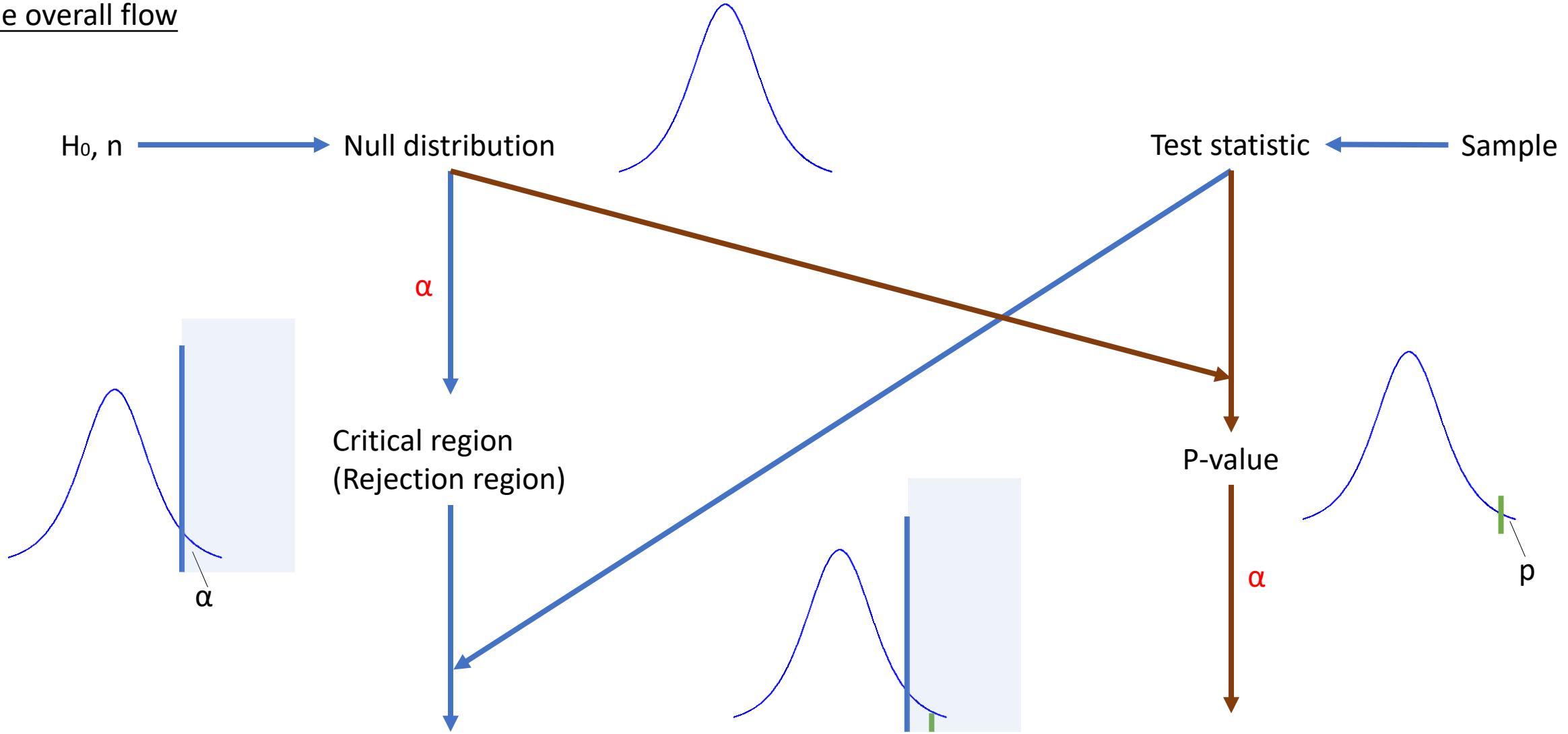
One-sample test of mean

Two-sample test of means

The overall flow



The overall flow



Test statistic falls into critical region → "Significant" → reject H_0

Test statistic falls outside critical region → "Insignificant" → accept H_0

$p \leq \alpha$ → "Significant" → reject H_0

$p > \alpha$ → "Insignificant" → accept H_0

The non-statistical things

Which hypothesis is the null and which is the alternative is not a mathematical one, and depends on scientific context, custom, and convenience.

> Assign the hypothesis that you would like to show to be true as the alternative: data can only support the alternative hypothesis but NOT the null.
>
> In scientific investigations, the null hypothesis is often an existing explanation that must be discredited based on (data) evidence, in order to demonstrate the presence of some physical phenomenon or effect given in the alternative hypothesis.
>
> Rejecting the null hypothesis often leads to action.
>
> The consequences of incorrectly rejecting one hypothesis may be graver than those of incorrectly rejecting the other.

The choice of α is essentially arbitrary, but is heavily influenced by custom. Small values, such as 0.01 and 0.05, are commonly used.

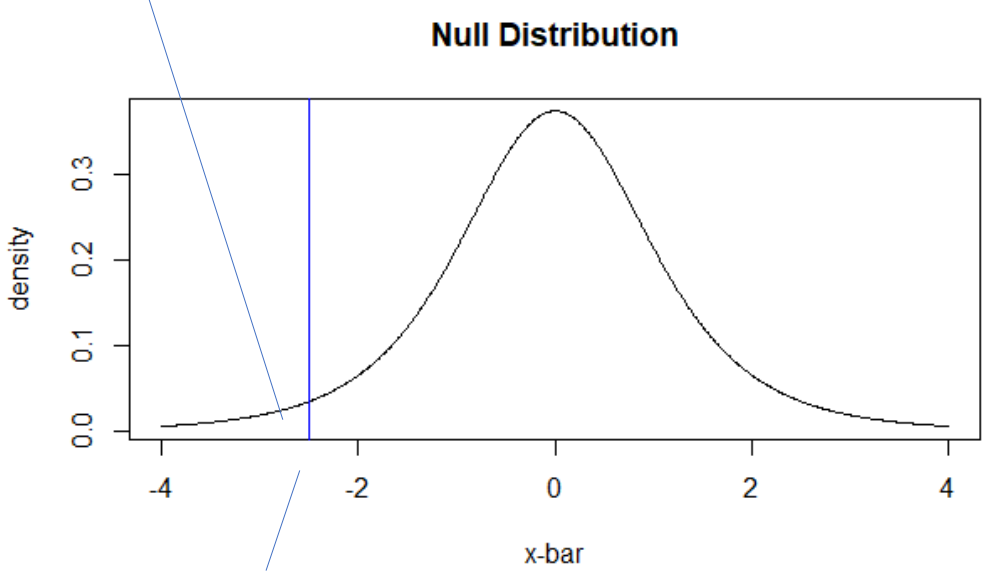
p-value

the smallest significance level at which the null hypothesis would be rejected

p-value is the probability of a result as or more extreme than that actually observed if the null hypothesis were true

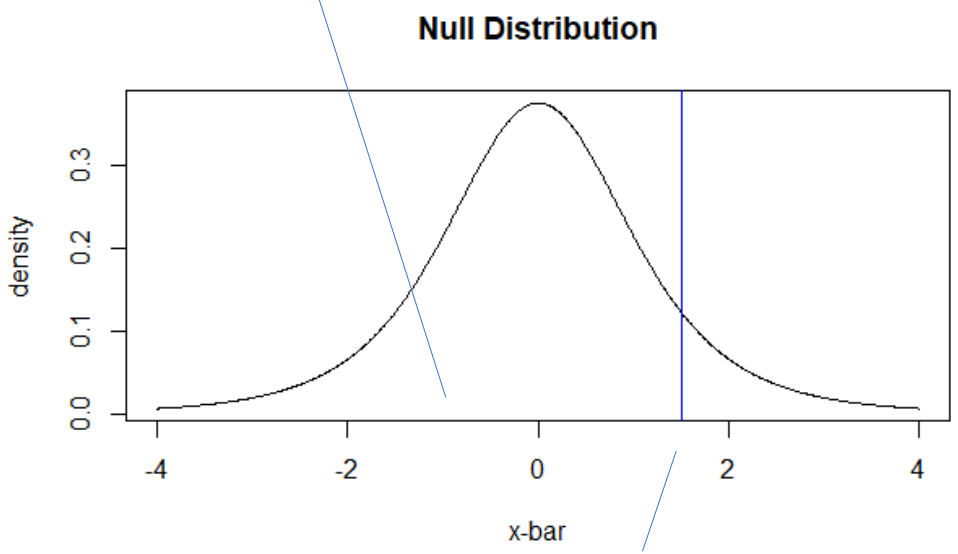
$H_0: \mu = \mu_0$
$H_1: \mu < \mu_0$

p=0.0334



$\bar{x} = -2.5$

p=0.896



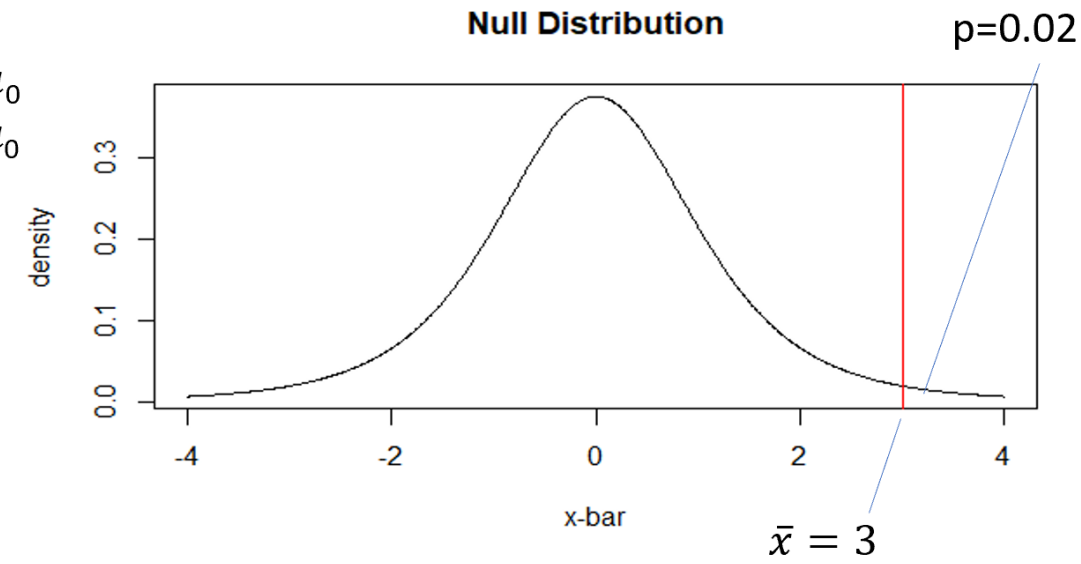
$\bar{x} = 1.5$

p-value

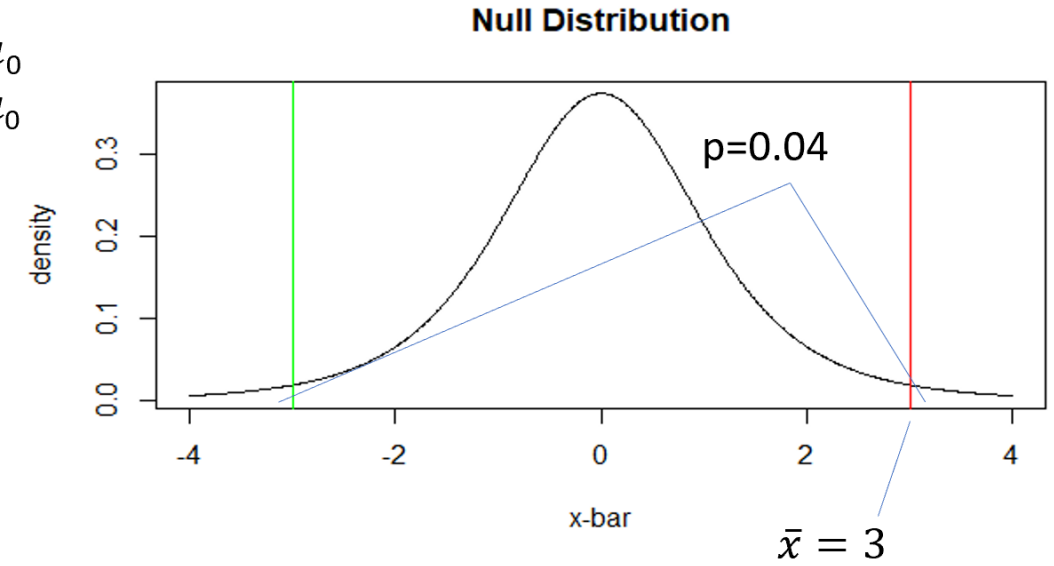
the smallest significance level at which the null hypothesis would be rejected

p-value is the probability of a result as or more extreme than that actually observed if the null hypothesis were true

$H_0: \mu = \mu_0$
$H_1: \mu > \mu_0$



$H_0: \mu = \mu_0$
$H_1: \mu \neq \mu_0$



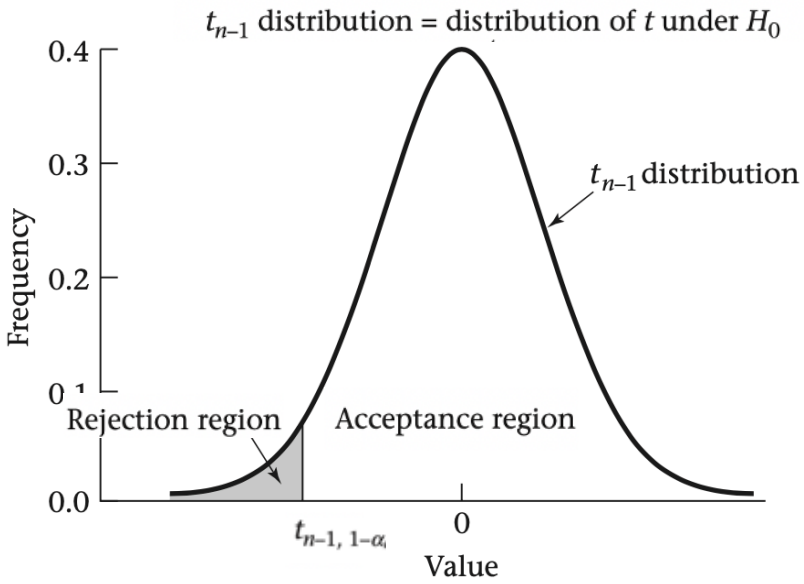
One sample test of mean; X is normally distributed, σ is unknown

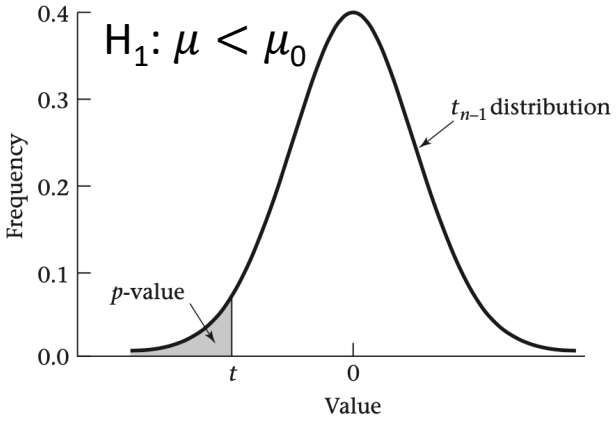
Distribution of test statistic under H_0	$T = \dfrac{\overline{X} - \mu}{S/\sqrt{n}}$	$\sim t_{n-1}$

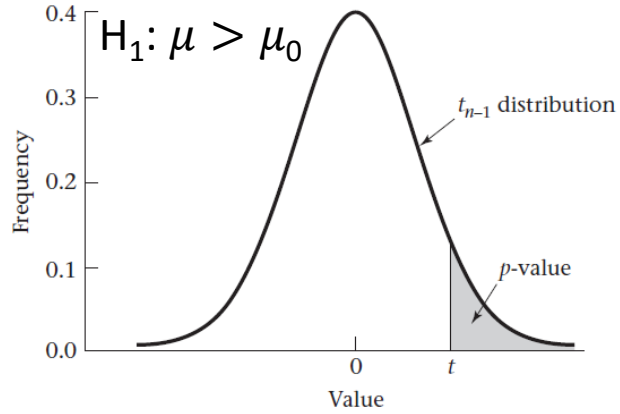
$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

One-sided alternative

$H_0: \mu = \mu_0$
$H_1: \mu < \mu_0$







When σ is known, replace s by σ, replace the t distribution by N(0,1) distribution

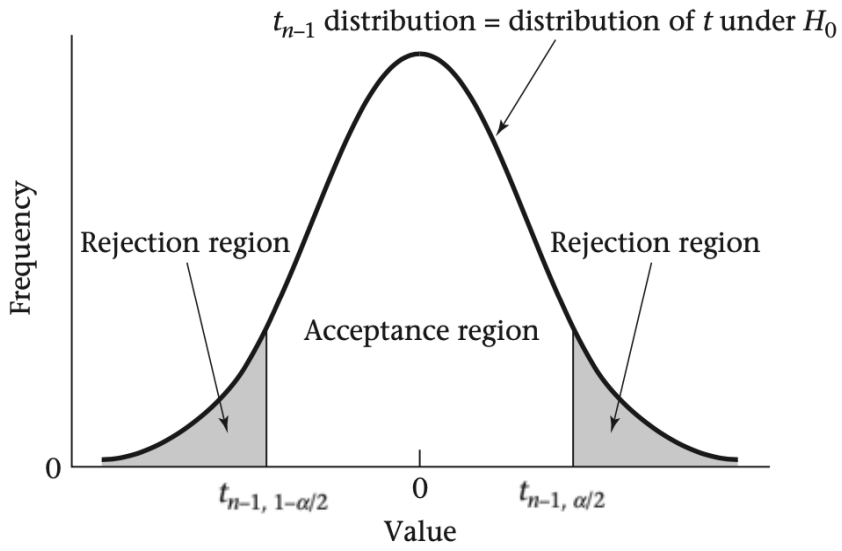
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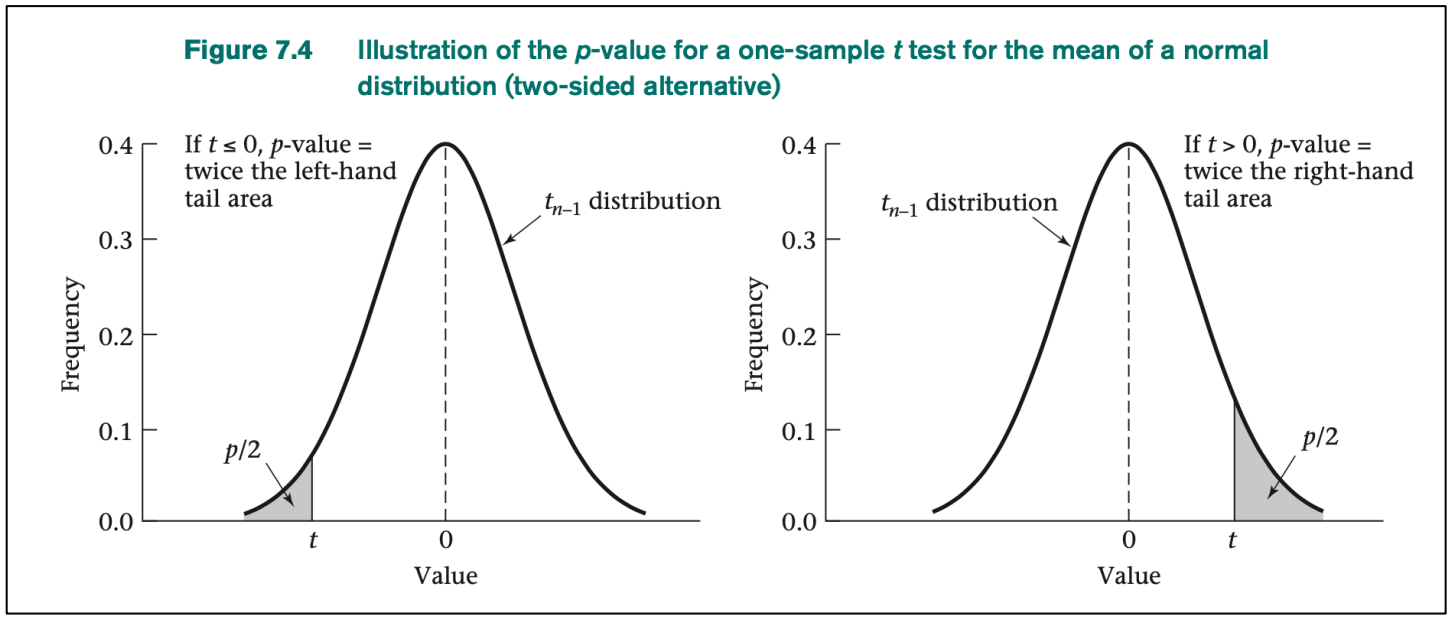
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$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

Two-sided alternative

$H_0: \mu = \mu_0$
$H_1: \mu \neq \mu_0$





When σ is known, replace s by σ, replace the t distribution by N(0,1) distribution

The Duality of Confidence Intervals and Hypothesis Tests

Let $X_1, \ldots, X_n$ be a random sample from a normal distribution having unknown mean μ and known variance σ^2. We consider testing the following hypotheses:

$$H_0: \mu = \mu_0$$
$$H_A: \mu \neq \mu_0$$

The acceptance region:

$$|\bar{x} - \mu_0| < z_{\alpha/2}\sigma/\sqrt{n}$$

A $100(1-\alpha)\%$ confidence interval for μ_0 is

$$\left[\overline{X} - z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right), \overline{X} + z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right)\right]$$

(When σ is unknown)

$$|\bar{x} - \mu_0| < t_{\alpha/2}(n-1)s/\sqrt{n}$$

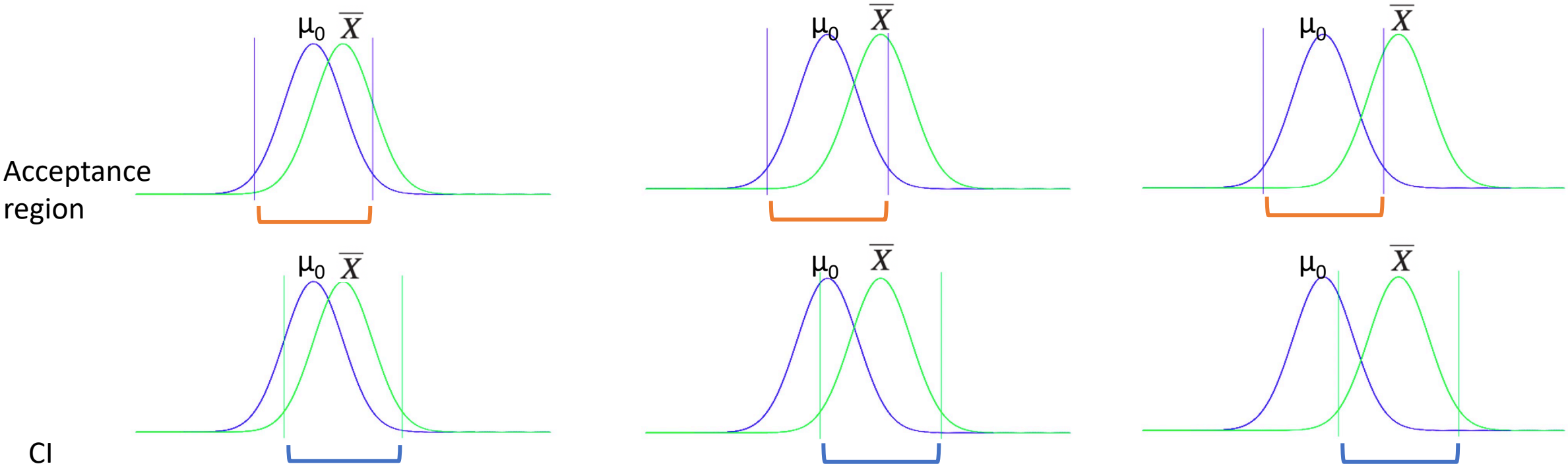
$$\left[\bar{x} - t_{\alpha/2}(n-1)\left(\frac{s}{\sqrt{n}}\right), \bar{x} + t_{\alpha/2}(n-1)\left(\frac{s}{\sqrt{n}}\right)\right]$$

μ_0 lies in the confidence interval for μ if and only if the hypothesis test accepts.

The confidence interval consists precisely of all those values of μ_0 *for which the null hypothesis* H_0: $\mu = \mu_0$ *is accepted*.

The Duality of Confidence Intervals and Hypothesis Tests

A symmetric relationship



It all boils down to the distance between μ_0 and $\overline{X}$

Example

A major city provides one of the most comprehensive emergency medical services in the world. Operating in a multiple hospital system with approximately 20 mobile medical units, the service goal is to respond to medical emergencies with a mean time of 12 minutes or less.

The director of medical services wants to formulate a hypothesis test that could use a sample of emergency response times to determine whether or not the service goal of 12 minutes or less is being achieved. **Suppose $n = 40$, $\overline{X} = 13.25$ minutes, $s = 3.2$ minutes, $\alpha = 0.05$**

One-Tailed Test about a Population Mean

$H_0: \mu \leq 12$ $\quad$ $H_1: \mu > 12$

$$t = \frac{\overline{x} - \mu}{s/\sqrt{n}} = \frac{13.25 - 12}{3.2/\sqrt{40}} = 2.47$$

$$t_{n-1,\alpha} = t_{39,.05} = 1.685$$

Since 2.47 > 1.685, we can reject H_0 at a significance level of .05.

$t_{n-1,\alpha} = qt(1-\alpha, n-1) = qt(0.95,39) = 1.685$

Note: the R function $qt(p, df)$ calculates the value from a t distribution with df degrees of freedom such that the area to its left is p.

Example

What if $\alpha = 0.005$?

$t_{39,\,0.005} = = qt(1-\alpha, n-1) = qt(0.995,39) = 2.707 > 2.47$

we fail to reject H_0 at significance level 0.005.

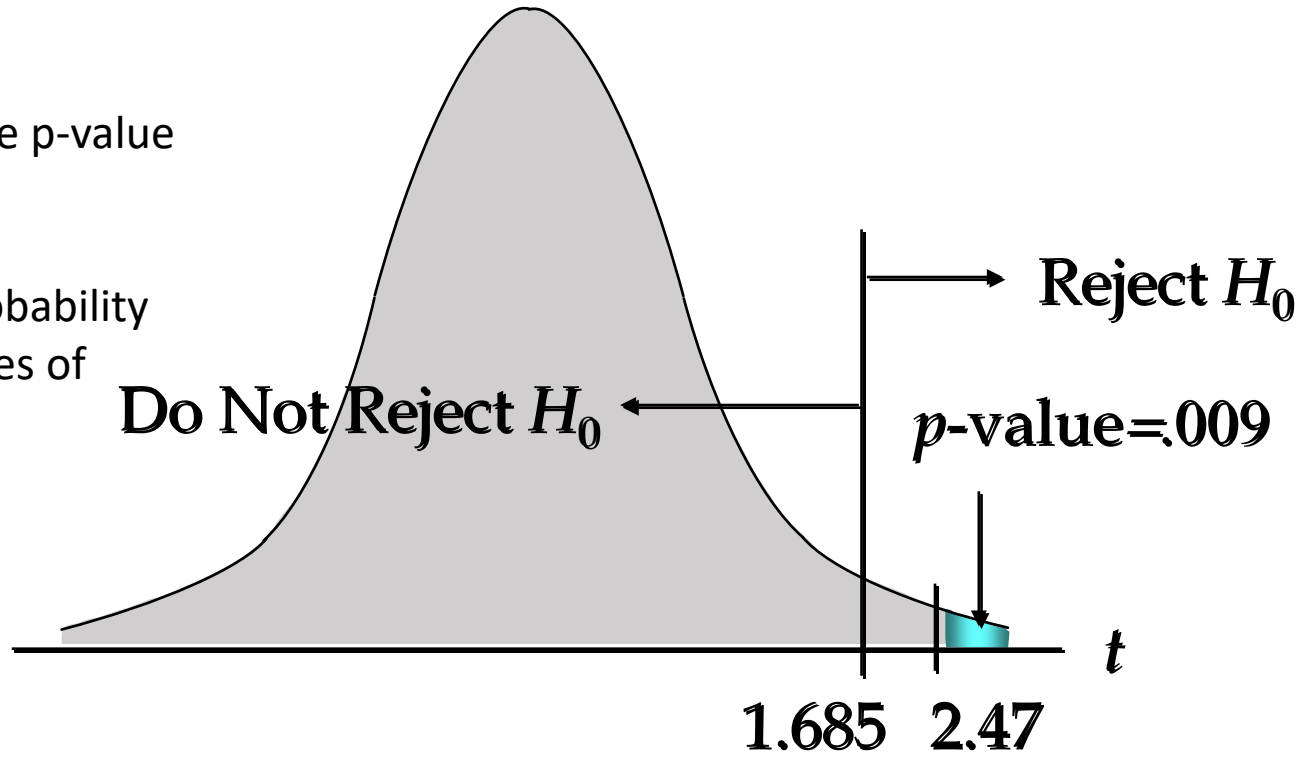
Using the *p*-value to Test the Hypothesis

Recall that $t = 2.47$ for $\overline{x} = 13.25$.

Pr ($t_{39} \geq 2.47$) = 1 - *pt*(2.47,39) = .009, which is the p-value

If *p*-value < α, we reject H_0.

Note: the function $pt(val, df)$ calculates the probability less than val from a *t* distribution with df degrees of freedom.



Example

An n increases, the standard error decreases.

It allows us to reject the null hypothesis even if the absolute difference is small – higher statistical power at a given simple alternative hypothesis.

Use the Metro EMS example again

Suppose the mean response time is 12.5, based on a sample of size 400.
Assess the result of the study.

The test statistic would be given by

$$\boldsymbol{t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{12.5 - 12}{3.2/\sqrt{400}} = 3.125}$$

Thus, the *p*-value is given by $Pr\,(t_{399} \geq 3.125) = 1 - pt(3.125,399) = .00095 < .05$
We reject H_0.

Suppose the mean response time is 13.5, based on a sample of size 10.
Assess the result of the study.

The test statistic would be given by

$$\boldsymbol{t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{13.5 - 12}{3.2/\sqrt{10}} = 1.482}$$

Thus, the *p*-value is given by $Pr\,(t_{9} \geq 1.482) = 1 - pt(1.482,9) = .086 > .05$
We fail to reject H_0.

Example: Glow Toothpaste

Two-Tailed Tests about a Population Mean

The production line for Glow toothpaste is designed to fill tubes of toothpaste with a mean weight of 6 ounces.

Periodically, a sample of 30 tubes will be selected in order to check the filling process. Quality assurance procedures call for the continuation of the filling process if the sample results are consistent with the assumption that the mean filling weight for the population of toothpaste tubes is 6 ounces; otherwise the filling process will be stopped and adjusted.

Hypotheses $H_0: \mu = 6 \quad H_1: \mu \neq 6$

Assuming a .05 level of significance, Reject H_0 if

$$|t| > t_{n-1,\alpha/2} = t_{29,0.05/2} = qt\left(1 - \frac{0.05}{2}, 29\right) = 2.045$$

Assume that a sample of 30 toothpaste tubes provides a sample mean of 6.1 ounces and standard deviation of 0.2 ounces.

Let n = 30, $\bar{x}$ = 6.1 ounces, s = .2 ounces

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{6.1 - 6}{.2/\sqrt{30}} = 2.74$$

Since 2.74 > 2.045, we reject H_0.

Conclusion: Under the 0.05 level of significance, we conclude that the mean filling weight of the toothpaste tubes is not 6 ounces. The filling process should be stopped and the filling mechanism adjusted.

Example: Glow Toothpaste

Two-Tailed Tests about a Population Mean

The production line for Glow toothpaste is designed to fill tubes of toothpaste with a mean weight of 6 ounces.

Periodically, a sample of 30 tubes will be selected in order to check the filling process. Quality assurance procedures call for the continuation of the filling process if the sample results are consistent with the assumption that the mean filling weight for the population of toothpaste tubes is 6 ounces; otherwise the filling process will be stopped and adjusted.

Assume that a sample of 30 toothpaste tubes provides a sample mean of 6.1 ounces and standard deviation of 0.2 ounces.

Let n = 30, $\bar{x}$ = 6.1 ounces, s = .2 ounces. $t = \frac{\bar{x}-\mu_0}{S/\sqrt{n}} = \frac{6.1-6}{.2/\sqrt{30}} = 2.74$

Using the *p*-Value

The p-value for a two-tailed hypothesis test is $2 \times Pr(t_{29} > 2.74)$

Given pt(2.74,29)=.995, we have

***p*-value = $2 \times Pr(t_{29} > 2.74)$ = 2 ×(1-*pt*(2.74,29))= 2(1-.995) = .01**

The *p*-value .01 is less than α = .05, so H_0 is rejected.

Example: Glow Toothpaste

Repeat the previous Glow Toothpaste example, use n =60

The production line for Glow toothpaste is designed to fill tubes of toothpaste with a mean weight of 6 ounces.

Periodically, a sample of **60** tubes will be selected in order to check the filling process. Quality assurance procedures call for the continuation of the filling process if the sample results are consistent with the assumption that the mean filling weight for the population of toothpaste tubes is 6 ounces; otherwise the filling process will be stopped and adjusted.

Hypotheses $H_0: \mu = 6 \quad H_1: \mu \neq 6$

Assuming a .05 level of significance, Reject H_0 if

$$|t| > t_{n-1,\alpha/2} = t_{59,0.05/2} = qt\left(1 - \frac{0.05}{2}, 59\right) = 2$$

Assume that a sample of 60 toothpaste tubes provides a sample mean of 6.1 ounces and standard deviation of 0.2 ounces.

Let n = 60, $\bar{x}$ = 6.1 ounces, s = .2 ounces

$$t = \frac{\bar{x} - \mu_0}{S/\sqrt{n}} = \frac{6.1 - 6}{.2/\sqrt{60}} = 3.87$$

Since 3.87 > 2, we reject H_0.

Conclusion: Under the 0.05 level of significance, we conclude that the mean filling weight of the toothpaste tubes is not 6 ounces. The filling process should be stopped and the filling mechanism adjusted.

Example: Glow Toothpaste

Repeat the previous Glow Toothpaste example, use n =60

The production line for Glow toothpaste is designed to fill tubes of toothpaste with a mean weight of 6 ounces.

Periodically, a sample of 60 tubes will be selected in order to check the filling process. Quality assurance procedures call for the continuation of the filling process if the sample results are consistent with the assumption that the mean filling weight for the population of toothpaste tubes is 6 ounces; otherwise the filling process will be stopped and adjusted.

Assume that a sample of 60 toothpaste tubes provides a sample mean of 6.1 ounces and standard deviation of 0.2 ounces.

Let n = 60, $\bar{x}$ = 6.1 ounces, s = .2 ounces. $t = \frac{\bar{x}-\mu_0}{S/\sqrt{n}} = \frac{6.1-6}{.2/\sqrt{60}} = 3.87$

Using the *p*-Value

The p-value for a two-tailed hypothesis test is $2 \times Pr(t_{59} > 3.87)$

***p*-value** = $2 \times Pr(t_{59} > 3.87)$ = 2(1-pt(3.87,59))= **.00027**

The p-value is less than α = .05, so H_0 is rejected.

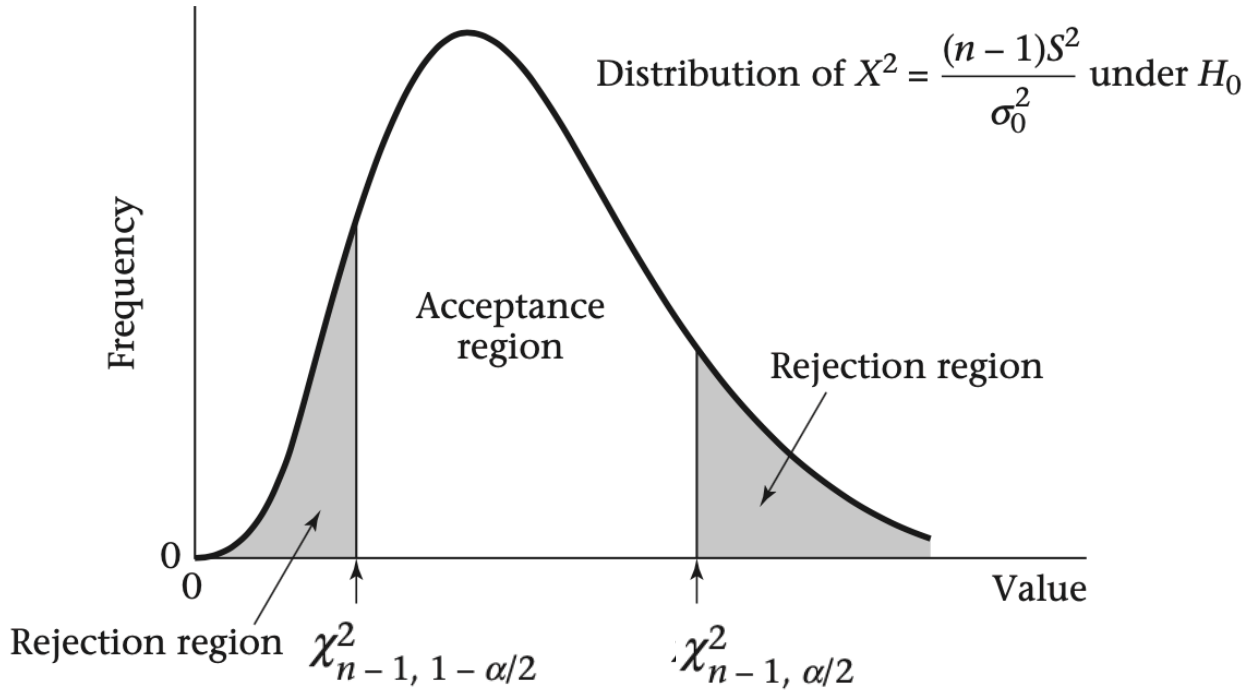
One sample test of variance

Distribution of test statistic under H_0

$$X^2 = \frac{(n-1)S^2}{\sigma_0^2} \sim \chi^2_{n-1}$$

$H_0: \sigma = \sigma_0$
$H_1: \sigma \neq \sigma_0$

Figure 7.10 **Acceptance and rejection regions for the one-sample χ^2 test for the variance of a normal distribution (two-sided alternative)**



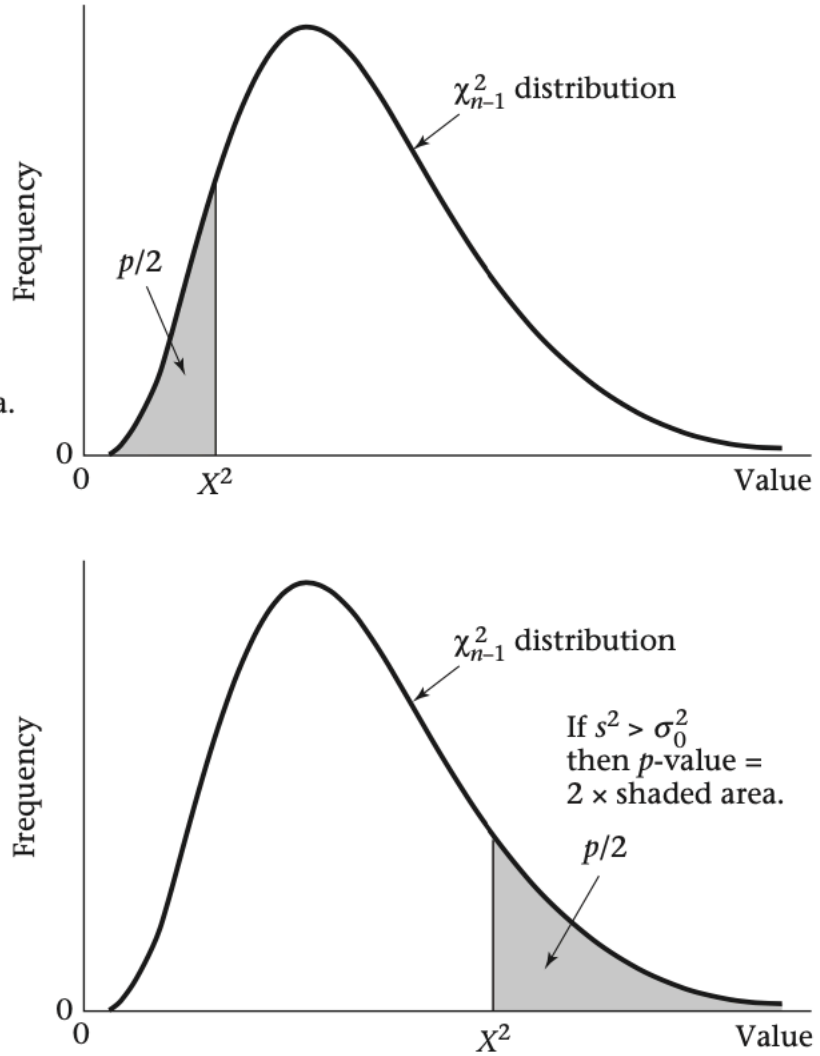
One sample test of variance

Let the test statistic $X^2 = \dfrac{(n-1)s^2}{\sigma_0^2}$.

If $s^2 \le \sigma_0^2$, then p-value = 2 × (area to the left of X^2 under a χ^2_{n-1} distribution).

If $s^2 > \sigma_0^2$, then p-value = 2 × (area to the right of X^2 under a χ^2_{n-1} distribution).

Figure 7.11 **Illustration of the p-value for a one-sample χ^2 test for the variance of a normal distribution (two-sided alternative)**



Example

Consider the Glow Toothpaste exmaple again with n = 30.

Suppose we want to test: H_0: $\sigma^2 = 0.25^2$ vs H_1: $\sigma^2 \neq 0.25^2$

We know tha n = 30, s = 0.2 ($s^2 = 0.2^2 = 0.04$), $\sigma_0 = 0.25$, we compute the test statistics X^2 given by

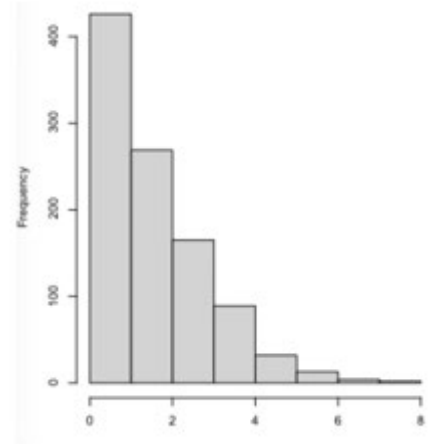
$$X^2 = \frac{(n-1)s^2}{{\sigma_0}^2} = \frac{29(0.2)^2}{(0.25)^2} = 18.56$$

p-value: $2 \times Pr({X^2}_{29} < 18.56) = 0.136$

In R, we use pchisq(val, df) to compute the cdf of a chisquare distribution with degree of freedom = df

```
> pchisq(18.56, 29) * 2
[1] 0.1358972
```

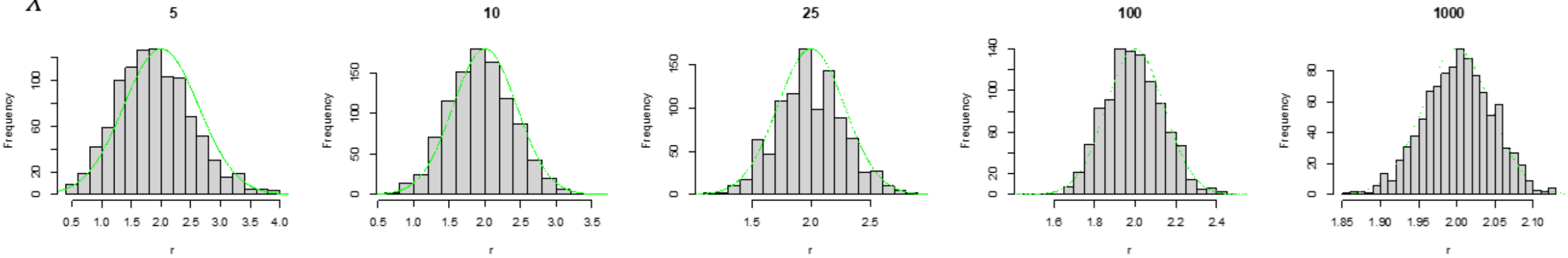
What about normality?



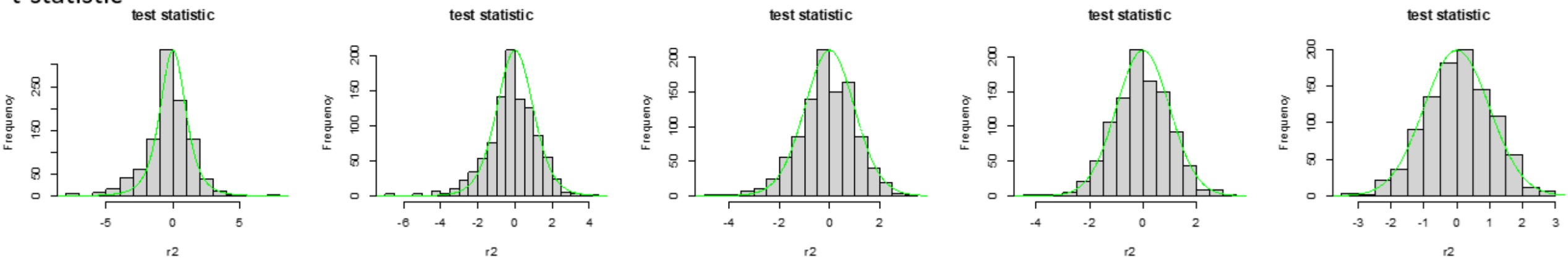
In many situations we don't know if X follows the normal distribution.

The t-test is robust against non-normality – the CLT says $\bar{X}$ approaches normal distribution when the sample size goes large.

$\bar{X}$



t-statistic



Two sample test of means

$H_0: \mu_1 = \mu_2$ vs. $H_1: \mu_1 \neq \mu_2$

$$\bar{x} - \bar{y} \sim N(\mu_x - \mu_y, \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m})$$

If m and n are not large (say, considerably smaller than 30), and if we can assume the two groups have the same variance

$$\sigma_X^2 = \sigma_Y^2 = \sigma^2$$

The pooled variance (to stabilize the variance estimate)

$$U = \frac{(n-1)S_X^2}{\sigma^2} + \frac{(m-1)S_Y^2}{\sigma^2} \quad \sim \quad \chi^2(n+m-2)$$

$$T = \frac{Z}{\sqrt{U/(n+m-2)}} \quad \sim \quad t_{n+m-2}$$

$$T = \frac{\dfrac{\overline{X} - \overline{Y} - (\mu_X - \mu_Y)}{\sqrt{\sigma^2/n + \sigma^2/m}}}{\sqrt{\left[\dfrac{(n-1)S_X^2}{\sigma^2} + \dfrac{(m-1)S_Y^2}{\sigma^2}\right] \Big/ (n+m-2)}} \quad \sim \quad t_{n+m-2}$$

Two sample test of means

Under the null:

$$t = \frac{\overline{X} - \overline{Y}}{\sqrt{\left[\frac{(n-1)S_X^2 + (m-1)S_Y^2}{n+m-2}\right]\left[\frac{1}{n} + \frac{1}{m}\right]}} \quad \sim \quad t_{n+m-2}$$

Reject the null if

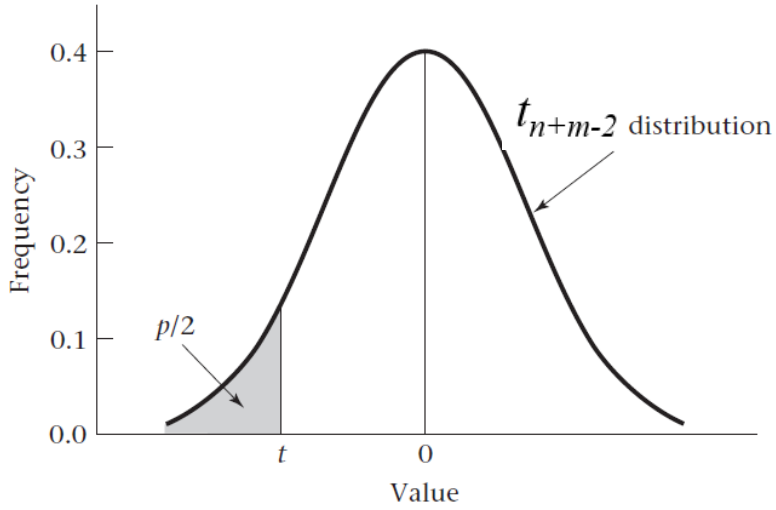
$$t > t_{n+m-2,\alpha/2}$$

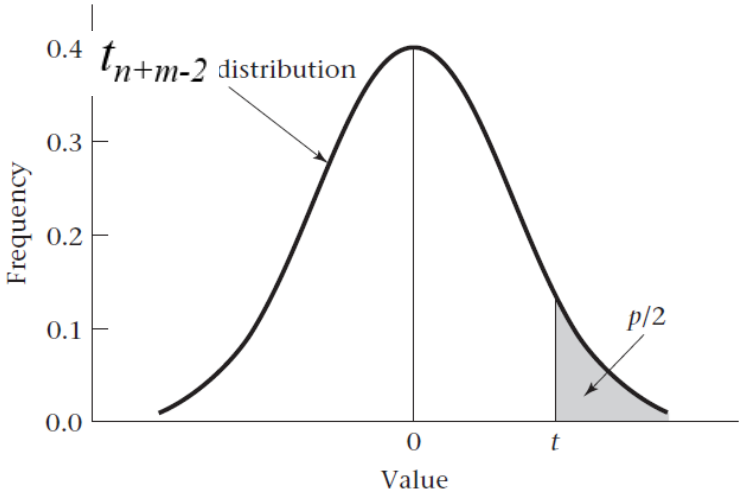
or

$$t < t_{n+m-2,1-\alpha/2} = -t_{n+m-2,\alpha/2}$$

If $t \le 0$, $p = 2 \times$ (area to the left of t under a $t_{n_1+n_2-2}$ distribution).

If $t > 0$, $p = 2 \times$ (area to the right of t under a $t_{n_1+n_2-2}$ distribution).





Two sample test of variance

But are the variances the same?

Reminder:

DEFINITION

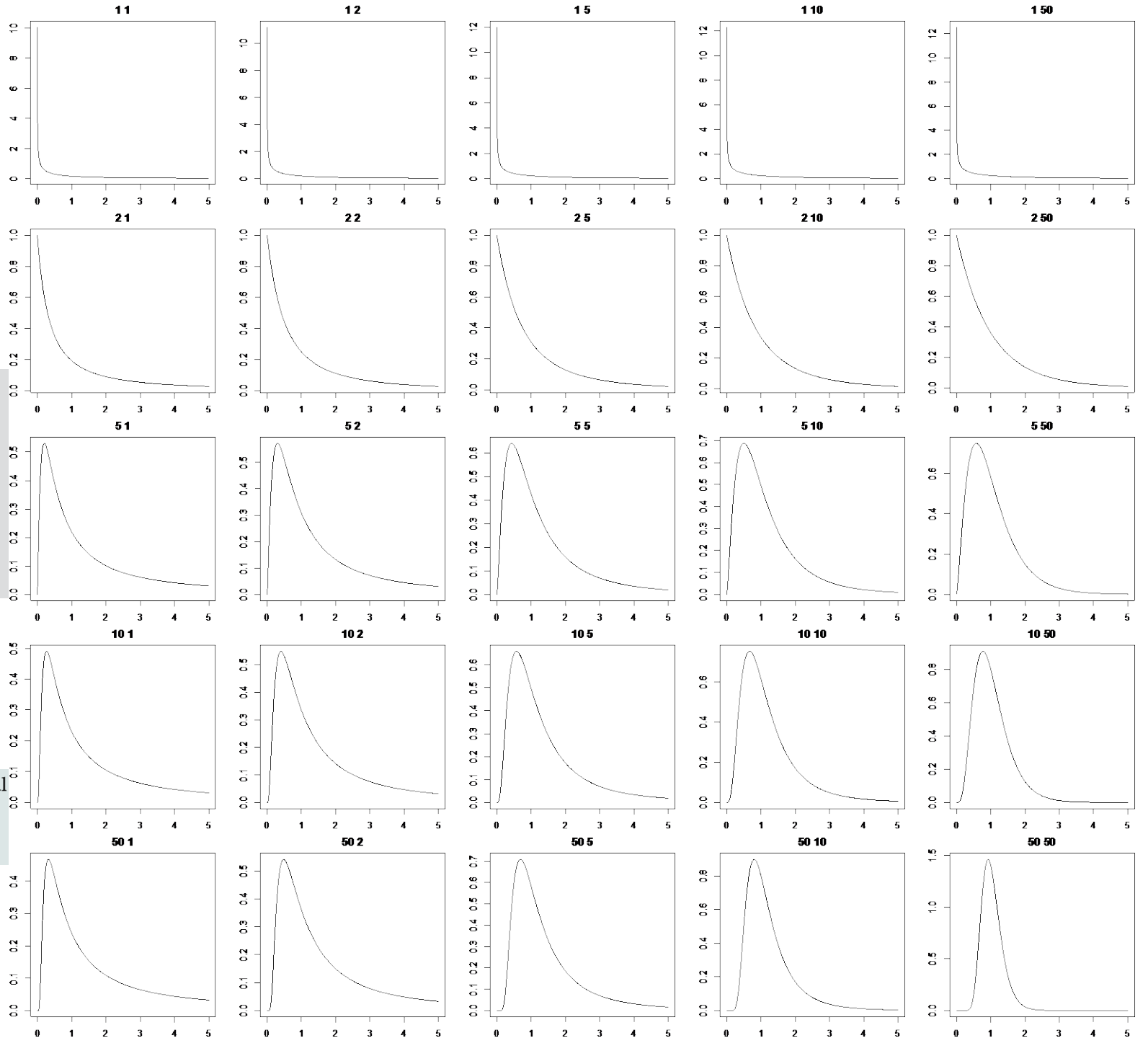
Let U and V be independent chi-square random variables with m and n degrees of freedom, respectively. The distribution of

$$W = \frac{U/m}{V/n}$$

is called the F distribution with m and n degrees of freedom and is denoted by $F_{m,n}$. ■

The **lower *p*th percentile** of an F distribution with d_1 and d_2 *df is* the reciprocal of the **upper *p*th percentile** of an F distribution with d_2 and d_1 *df*. In symbols,

$$F_{d_1,d_2,p} = 1/F_{d_2,d_1,1-p}$$



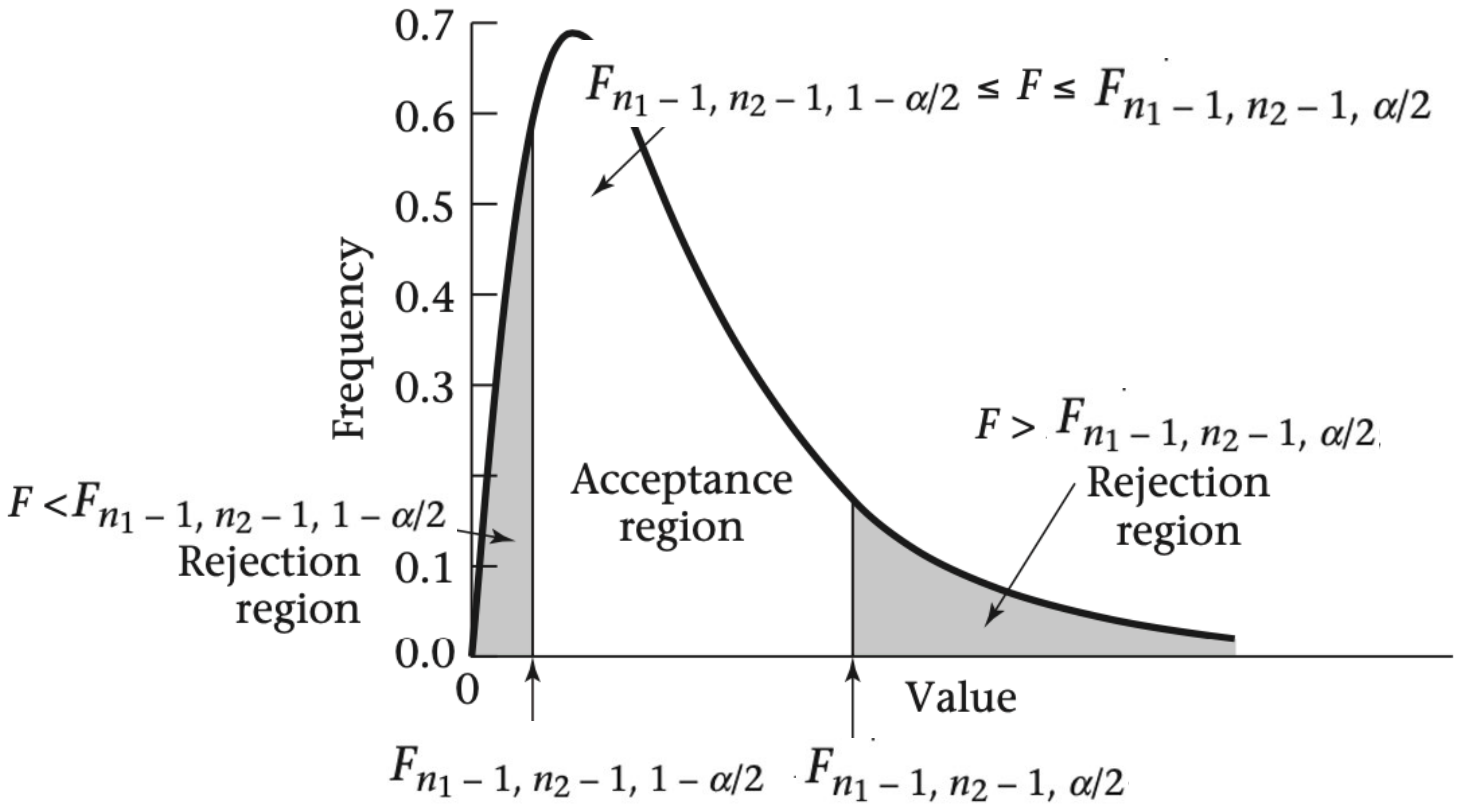
The distribution of $(n-1)S^2/\sigma^2$ is the chi-square distribution with $n-1$ degrees of freedom.

Two sample test of variance

$H_0: \sigma_1^2 = \sigma_2^2$

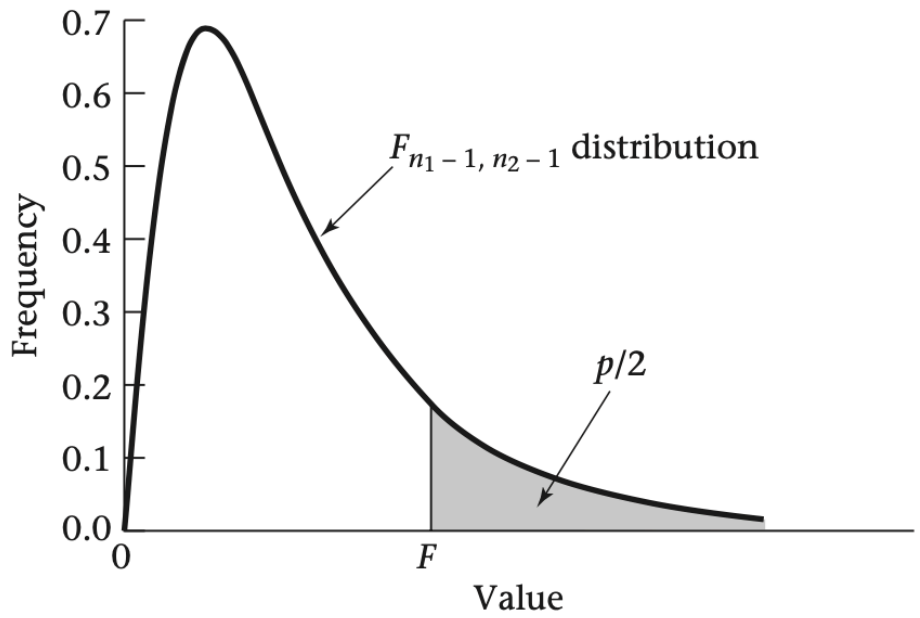
$H_1: \sigma_1^2 \neq \sigma_2^2$

test statistic $F = s_1^2/s_2^2$.

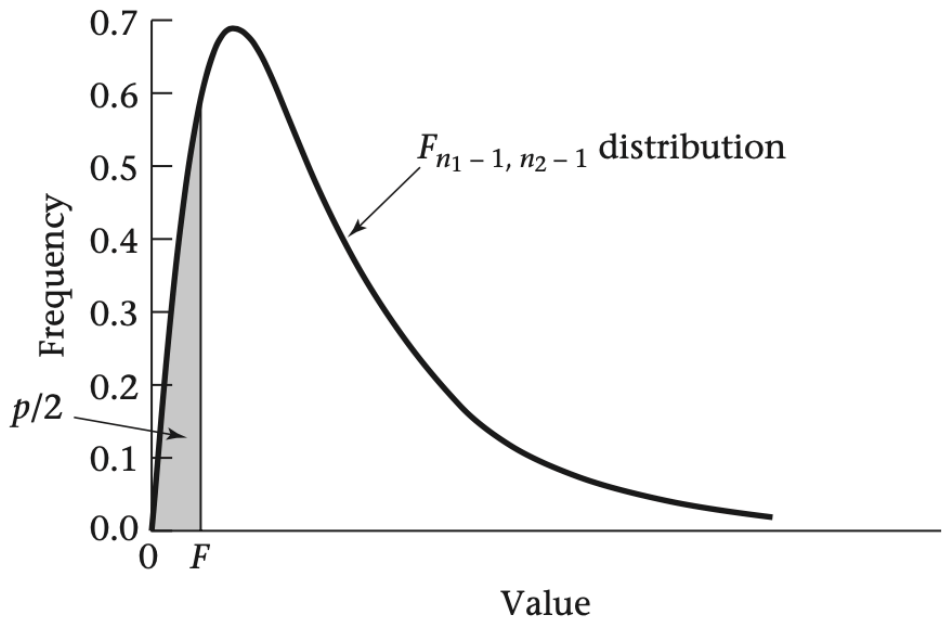


$F_{n_1-1,\,n_2-1}$ distribution = distribution of $F = S_1^2/S_2^2$ under H_0

Two sample test of variance



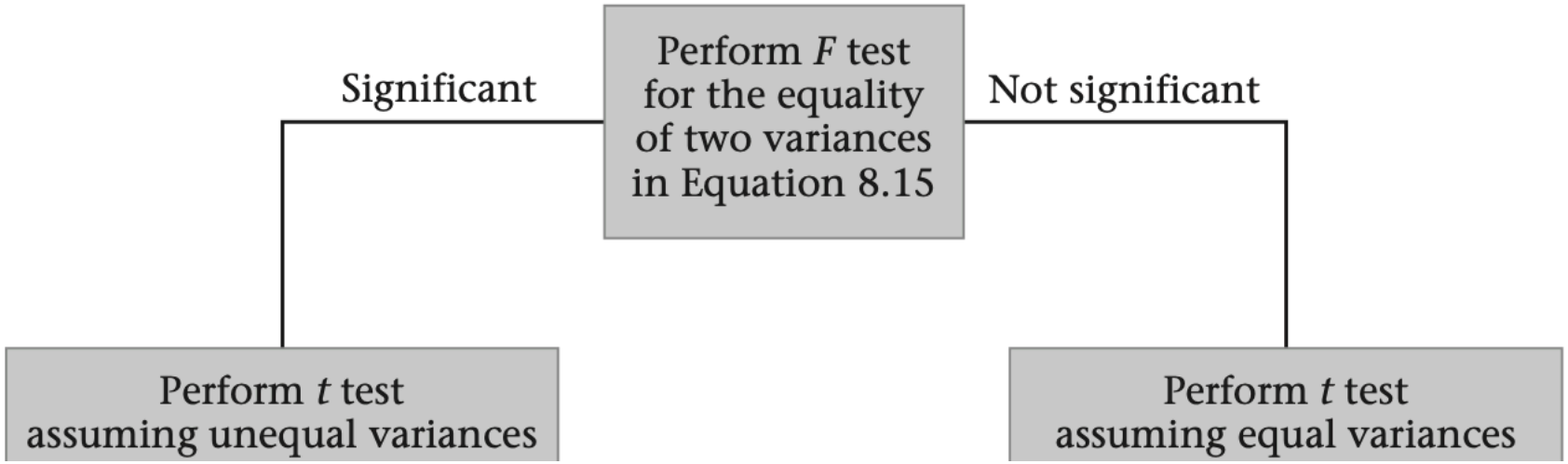
If $F = s_1^2/s_2^2 \geq 1$, then $p = 2 \times$ (area to the right of F under an $F_{n_1-1,\,n_2-1}$ distribution)



If $F = s_1^2/s_2^2 < 1$, then $p = 2 \times$ (area to the left of F under an $F_{n_1-1,\,n_2-1}$ distribution)

Two sample test of variance

Testing for equality of means between two independent samples



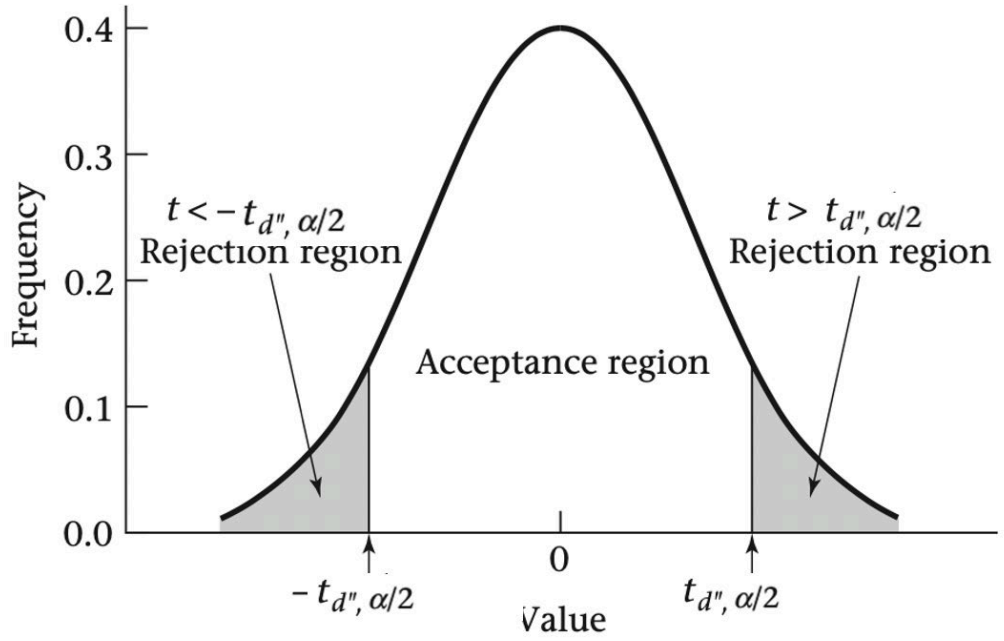
Two sample test of means (unequal variances)

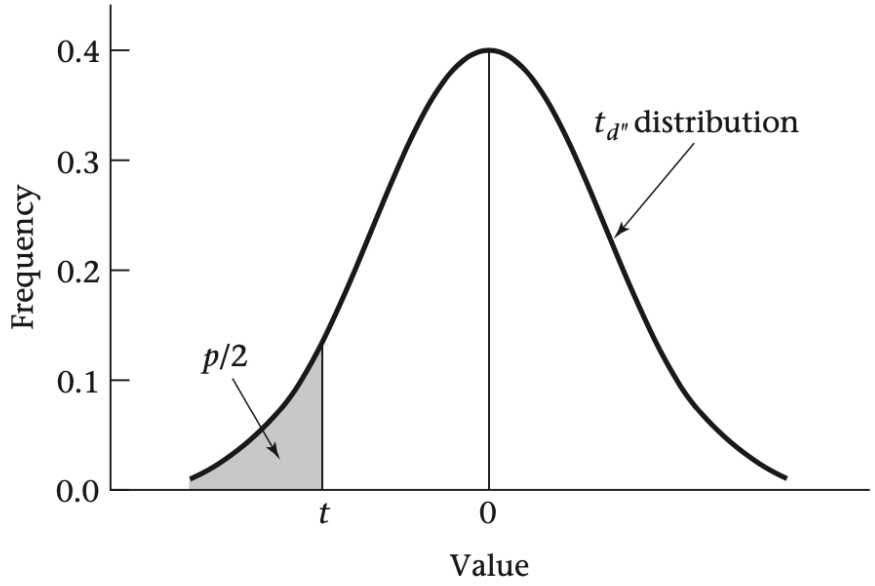
(1) Compute the test statistic

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

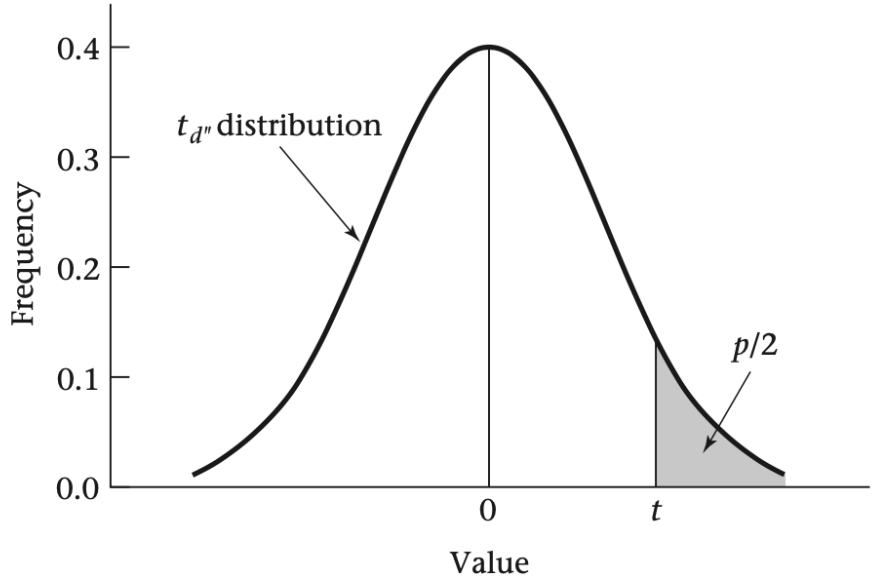
(2) Compute the approximate degrees of freedom d', where

$$d' = \frac{\left(s_1^2/n_1 + s_2^2/n_2\right)^2}{\left(s_1^2/n_1\right)^2/(n_1 - 1) + \left(s_2^2/n_2\right)^2/(n_2 - 1)}$$





If $t = (\bar{x}_1 - \bar{x}_2)/\sqrt{s_1^2/n_1 + s_2^2/n_2} \le 0$, then $p = 2 \times$ (area to the left of t under a $t_{d''}$ distribution)

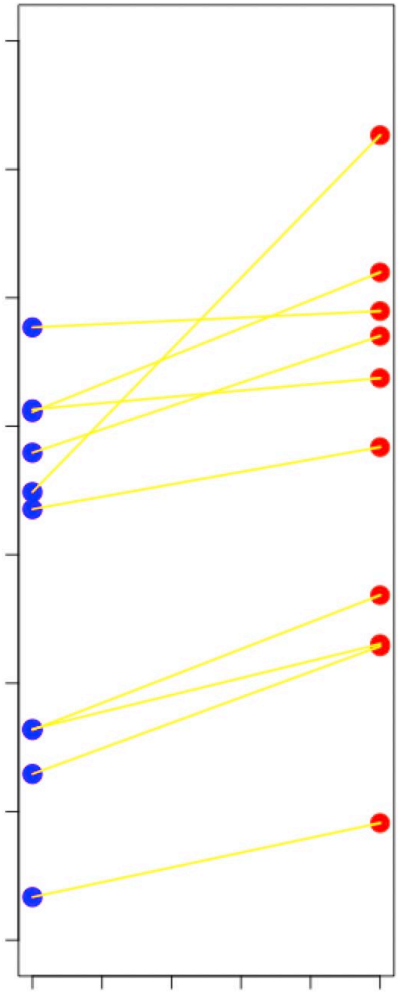


If $t = (\bar{x}_1 - \bar{x}_2)/\sqrt{s_1^2/n_1 + s_2^2/n_2} > 0$, then $p = 2 \times$ (area to the right of t under a $t_{d''}$ distribution)

Paired samples

Definition 8.4 Two samples are said to be **paired** when each data point in the first sample is matched and is related to a unique data point in the second sample.

Example 8.3 The paired samples may represent two sets of measurements on the same people. In this case each person is serving as his or her own control, as in Equation 8.1. The paired samples may also represent measurements on different people who are chosen on an individual basis using matching criteria, such as age and sex, to be very similar to each other.



In some situations, the interest is the **mean of the within-pair difference**, but NOT the difference in means of the two groups.

First take difference within pair, then treat the difference as a single RV.

Conduct a one-sample t test based on the differences (d_i).

Examples of two-sample comparison

Hypertension Suppose a sample of eight 35- to 39-year-old nonprenant, premenopausal OC users who work in a company and have a mean systolic blood pressure (SBP) of 132.86 mm Hg and sample standard deviation of 15.34 mm Hg are identified. A sample of 21 nonpregnant, premenopausal, non-OC users in the same age group are similarly identified who have mean SBP of 127.44 mm Hg and sample standard deviation of 18.23 mm Hg. What can be said about the underlying mean difference in blood pressure between the two groups?

We want to test the hypothesis $H_0: \mu_1 = \mu_2$ $H_1: \mu_1 \neq \mu_2$

The common variance is first estimated:

$$s^2 = \frac{7(15.34)^2 + 20(18.23)^2}{27} = \frac{8293.9}{27} = 307.18$$

or $s = 17.527$. The following test statistic is then computed:

$$t = \frac{132.86 - 127.44}{17.527\sqrt{1/8 + 1/21}} = \frac{5.42}{17.527 \times 0.415} = \frac{5.42}{7.282} = 0.74$$

$\alpha = 0.05$ $t_{27,0.025} = qt(1 - 0.025, 27) = 2.052$

Since $-2.052 \leq 0.74 \leq 2.052$, we fail to reject H_0.

Examples of two-sample comparison

Hypertension Test whether the variance of blood pressure is significantly different between OC users and non-OC users.
The sample standard deviation of blood pressure for the 8 OC users was 15.34 and for the 21 non-OC users was 18.23. Hence the variance ratio is

$$F = (\frac{15.34}{18.23})^2 = 0.708$$

$$\Pr(F_{7,20} \leq 0.708) = 0.334$$

The two–tailed p-value = $2 \times \Pr(F_{7,20} \leq 0.708) = 2 \times pf(0.708,7,20) = 2 * 0.334 = 0.668$

In R, this is obtained using the *pf*() function: *pf(F,* df_1,df_2) is Pr($F_{df1,df2} \leq F$)

Examples of two-sample comparison

Cardiovascular Disease, Pediatrics Consider a problem studying the familial aggregation of cholesterol levels. In particular, suppose cholesterol levels are assessed in 100 children, 2 to 14 years of age, of men who have died from heart disease and it is found that the mean cholesterol level in the group ($\bar{x}$) is 207.3 mg/dL. Suppose the sample standard deviation in this group (s_1) is 35.6 mg/dL.

Among 74 control children, the mean cholesterol level ($\bar{x}$) is 193.4 mg/dL with a sample standard deviation (s_2) of 17.3 mg/dL. We would like to compare the means of these two groups using the two-sample t test for independent samples

We first test: $H_0: \sigma_1^2 = \sigma_2^2$ vs. $H_0: \sigma_1^2 \neq \sigma_2^2$

$$F = \frac{35.6^2}{17.3^2} = 4.23$$

The two-tailed p-value is $2 \times \Pr(F_{99,73} > 4.23) = 2 * (1 - pf(4.23, 99,73)) \approx 0$

In R:

```
> 2*(1-pf(4.23,99,73))
[1] 8.839516e-10
```

Examples of two-sample comparison

Cardiovascular Disease, Pediatrics Consider a problem studying the familial aggregation of cholesterol levels. In particular, suppose cholesterol levels are assessed in 100 children, 2 to 14 years of age, of men who have died from heart disease and it is found that the mean cholesterol level in the group ($\bar{x}_1$) is 207.3 mg/dL. Suppose the sample standard deviation in this group (s_1) is 35.6 mg/dL.

Among 74 control children, the mean cholesterol level ($\bar{x}_2$) is 193.4 mg/dL with a sample standard deviation (s_2) of 17.3 mg/dL. We would like to compare the means of these two groups using the two-sample t test for independent samples

$$t = \frac{207.3 - 193.4}{\sqrt{\frac{35.6^2}{100} + \frac{17.3^2}{74}}} = \frac{13.9}{4.089} = 3.40$$

$t = 3.40 > t_{151,0.025} = 1.976$

Use R to find $t_{151,0.025}$

```
> qt(0.975,151)
[1] 1.975799
```

$$d' = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_1}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{(n_1-1)} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{(n_2-1)}} = \frac{\left(\frac{35.6^2}{100} + \frac{17.3^2}{74}\right)^2}{\frac{\left(\frac{35.6^2}{100}\right)^2}{99} + \frac{\left(\frac{17.3^2}{74}\right)^2}{73}} = \frac{16.718^2}{1.8465} = 151.4$$